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Dual pinion and ring architecture for gear systems

Abstract

Gear systems include a dual pinion and ring architecture. A gear system includes a housing with a shaft extending in the housing. The shaft includes an eccentric section and is rotatable about an axis. The eccentric section has a center that circles around the axis when the shaft rotates. A pinion gear unit is mounted on the eccentric section and has two sets of gear teeth. A ring gear is fixed to the housing and engages with one set of the gear teeth. Another ring gear engages with the other set of gear teeth and is fixed to a rotatable member. The gear system operates so that rotation of the shaft at a first speed effects rotation of the rotatable member at a second speed through the shaft, the pinion gear unit, and the ring gears.

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Background/Summary

INTRODUCTION

(1) The technical field generally relates to gear systems, and more particularly relates to gear systems for transferring and multiplying torque from an input to an output through a compact and high-ratio capable dual pinion and ring architecture.

(2) Gear systems are employed in a wide range of manufactured products such as equipment, machines, tools, vehicles and other apparatus. Various types of gears may be employed and those gears may have a wide variety of gear teeth arrangements. Many gear systems transfer power between an input and an output while providing the ability to select speed ratios and torque multiplication ratios. The individual gears and their linkages may be assembled in some applications, such as a transmission system, that include various torque transfer mechanisms, including those such as brakes and clutches, that may be selectively engaged and disengaged to control the torque path.

(3) Considerations in the design of gear systems, in addition to the power, speed and torque that will be transmitted include weight, efficiency and complexity. Generally, systems that employ a greater number of individual gears require more manufacturing resources and result in greater weight. In addition, a higher number of gear meshes tends to decrease efficiency such as through spin losses. Other considerations include physical size, which impacts the ability to package the gears system in available space.

(4) Accordingly, there is an ongoing desire for gear systems that deliver desirable power, torque and speed requirements while doing so in an efficient and low mass manner. There is also a desire to provide high-ratio torque transfer in a compact package. Furthermore, other desirable features

and characteristics of the present disclosure will become apparent from the subsequent detailed description and the appended claims, taken in conjunction with the accompanying drawings and the foregoing introduction.

SUMMARY

(5) Gear systems include a dual pinion and ring architecture. In a number of embodiments, a gear system includes a housing with a shaft extending in the housing. The shaft includes an eccentric section and is rotatable about an axis. The eccentric section has a center that circles around the axis when the shaft rotates. A pinion gear unit is mounted on the eccentric section and has two sets of gear teeth. A ring gear is fixed to the housing and engages with one set of the gear teeth. Another ring gear engages with the other set of gear teeth and is fixed to a rotatable member. The gear system operates so that rotation of the shaft at a first speed effects rotation of the rotatable member at a second speed through the shaft, the pinion gear unit, and the ring gears.

(6) In additional embodiments, at least one counterweight extends from the shaft to balance the eccentric section.

(7) In additional embodiments, a bearing assembly is disposed between the rotatable member and the housing. Axial loads on the additional ring gear are transferred to the housing through the rotatable member and the bearing assembly.

(8) In additional embodiments, the pinion gear unit includes a stepped pinion gear.

(9) In additional embodiments, a bearing assembly is disposed between the eccentric section and the pinion gear unit so that the pinion gear unit is rotatable on the eccentric section.

(10) In additional embodiments, a counterweight extends from the shaft to balance the eccentric section. The output member covers a radial outside of the counterweight.

(11) In additional embodiments, the shaft includes a pair of bearing sections that are rotatably supported on the housing. The bearing sections and the ring gears are disposed about the axis coaxially and the eccentric section is disposed offset from the axis.

(12) In additional embodiments, one ring gear is larger in diameter than the other ring gear.

(13) In additional embodiments, no more than two meshing gear pairs are included. One meshing gear pair includes the pinion gear unit and a ring gear and another meshing gear pair includes the pinion gear unit and the other ring gear.

(14) In additional embodiments, the pinion gear unit includes a stepped pinion gear with a first pinion gear and a second pinion gear. The first pinion gear and the second pinion gear are configured to rotate and to orbit about the axis together on the eccentric section.

(15) In a number of additional embodiments, a gear system includes a housing and a gear set in the housing. The gear set includes a shaft extending in the housing. The shaft includes a base shaft section and an eccentric section. The shaft is rotatable about an axis, where the base shaft section is centered on the axis, and where the eccentric section has a center that is offset from the axis. The center circles around the axis when the shaft rotates. A pinion gear unit is mounted on the eccentric section. The pinion gear unit has a first pinion gear with a first set of gear teeth and a second pinion gear with a second set of gear teeth. A first ring gear engages with the first set of gear teeth. The first ring gear is fixed to the housing. A second ring gear engages with the second set of gear teeth. A rotatable member is fixed to the second ring gear. Rotation of the shaft at one speed effects rotation of the rotatable member at another speed through the shaft, the pinion gear unit, the first ring gear and the second ring gear.

(16) In additional embodiments, a pair of counterweights extend from the shaft to balance the eccentric section.

(17) In additional embodiments, a first bearing assembly is disposed between the rotatable member and the housing and a second bearing assembly disposed between the rotatable member and the shaft. Radial loads on the shaft are transferred to the housing through the second bearing assembly, the rotatable member and the first bearing assembly.

(18) In additional embodiments, the pinion gear unit includes a stepped pinion gear with the first

pinion gear and the second pinion gear formed as a unit.

(19) In additional embodiments, a needle-type bearing assembly is disposed between the eccentric section and the pinion gear unit so that the pinion gear unit is rotatable on the eccentric section.

(20) In additional embodiments, a pair of counterweights extend from the shaft to balance the eccentric section. The output member includes a flange that covers a radial outside of one counterweight of the pair of counterweights, and the output member includes an output shaft.

(21) In additional embodiments, the shaft includes a pair of bearing sections as part of the base shaft section so that the shaft is rotatably supported on the housing. The base shaft section, the first ring gear and the second ring gear are disposed about the axis coaxially and the eccentric section is disposed on a center that is offset from the axis.

(22) In additional embodiments, the first ring gear is larger in diameter than the second ring gear and the first pinion gear is larger than the second pinion gear.

(23) In additional embodiments, no more than two meshing gear pairs are included where one meshing gear pair includes the first pinion gear and the first ring gear and another meshing gear pair includes the second pinion gear and the second ring gear.

(24) In a number of other embodiments, a gear system for a vehicle that has a prime mover and an axle includes a housing fixed to the vehicle and a gear set in the housing. The gear set includes a shaft that is coupled with the prime mover and that extends in the housing. The shaft includes a base shaft section and an eccentric section. The shaft is rotatable about an axis. The base shaft section is centered on the axis and the eccentric section has a center that is offset from the axis. The center circles around the axis when the shaft rotates. A pinion gear unit is mounted on the eccentric section. The pinion gear unit has a first pinion gear with a first set of gear teeth and a second pinion gear with a second set of gear teeth. A first ring gear engages with the first set of gear teeth. The first ring gear is fixed to the housing. A second ring gear engages with the second set of gear teeth. A rotatable member is coupled with the axle and fixed to the second ring gear. The gear system is configured so that rotation of the shaft by the prime mover at a first speed effects rotation of the rotatable member and the axle at a second speed through the shaft, the pinion gear unit, the first ring gear and the second ring gear. The first pinion gear and the second pinion gear are configured to rotate and to orbit about the axis together on the eccentric section.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The exemplary embodiments will hereinafter be described in conjunction with the following drawing figures, wherein like numerals denote like elements, and wherein:

(2) FIG. 1 is a functional block diagram of a vehicle including a dual pinion and ring architecture gear system, in accordance with various embodiments;

(3) FIG. 2 is a schematic diagram of the dual pinion and ring architecture gear system of the vehicle of FIG. 1, in accordance with various embodiments;

(4) FIG. 3 is a sectional perspective view of aspects of the gearset exhibiting the dual pinion and ring architecture of FIG. 2 and shown removed from its application, in accordance with various embodiments;

(5) FIG. 4 is a perspective view of the shaft, eccentric and counterweights of the gearset of FIG. 3, shown in isolation, in accordance with various embodiments; and

(6) FIG. 5 is a perspective view of the pinion gear unit of the gearset of FIG. 3, shown in isolation, in accordance with various embodiments.

DETAILED DESCRIPTION

(7) The following detailed description is merely exemplary in nature and is not intended to limit the application and uses. Furthermore, there is no intention to be bound by any expressed or implied

theory presented in the preceding introduction or the following detailed description.

(8) FIG. 1 illustrates a vehicle **100**, according to an exemplary embodiment. As described in greater detail further below, the vehicle **100** includes powertrain **102** with a prime mover **104**. A gear system **106** is installed in the powertrain **102** but may be used at any location in the vehicle **100** or in another application where power transfer is employed. For example, in other embodiments the gear system **106** may be included in any of various types of machinery and equipment.

(9) In certain embodiments, the vehicle **100** includes a front axle **114** and a rear axle **116**. In some embodiments, the vehicle **100** comprises an automobile. In various embodiments, the vehicle **100** may be any one of a number of different types of automobiles, such as, for example, a sedan, a wagon, a truck, or a sport utility vehicle (SUV), and may be two-wheel drive (2WD) (i.e., rear-wheel drive or front-wheel drive), four-wheel drive (4WD) or all-wheel drive (AWD), and/or various other types of vehicles in certain embodiments. In certain embodiments, the vehicle **100** may also comprise one or more other types of vehicles. In addition, in various embodiments, it will also be appreciated that the vehicle **100** may comprise any number of other types of mobile platforms.

(10) In the depicted embodiment, the body **110** of the vehicle **100** substantially encloses other components of the vehicle **100**. Also in the depicted embodiment, the vehicle **100** includes the above-referenced axles **114**, **116** and front wheels **112**, as well as rear wheels **115**. The wheels **112**, **115** are each rotationally coupled to a respective one or more of the axles **114**, **116** near a respective corner of the body **110** to facilitate movement of the vehicle **100**. In one embodiment, the vehicle **100** includes four wheels **112**, **115** (namely, two front wheels **112** and two rear wheels **115**), although this may vary in other embodiments (for example for trucks and certain other vehicles).

(11) In various embodiments, powertrain **102** drives the wheels **112** and/or **115**. The powertrain **102** preferably comprises a propulsion system. In certain exemplary embodiments, the powertrain comprises an internal combustion engine and/or an electric motor/generator as the prime mover **104**. In certain embodiments, the powertrain **102** may vary, and/or two or more prime mover **104** types may be used. By way of example, the vehicle **100** may also incorporate any one of, or combination of, a number of different types of powertrains **102**, such as, for example, a gasoline or diesel fueled combustion engine, a “flex fuel vehicle” (FFV) engine (i.e., using a mixture of gasoline and alcohol), a gaseous compound (e.g., hydrogen and/or natural gas) fueled engine, a combustion/electric motor hybrid engine, and an electric motor. In the current embodiment, the prime mover **104** is an electric motor.

(12) The gear system **106** includes a dual pinion and ring architecture as described in more detail below, and is coupled between the prime mover **104** and the axle **114**. Through the axle **114**, the prime mover **104** is coupled with the wheels **112** for propelling the vehicle **100**. The coupling may be effected through a variety of arrangements such as to accommodate a transverse or longitudinally disposed prime mover **104** and may include a differential or other mechanism to connect with one or two half shafts. The architecture of the gear system **106** is adapted to applications such as in an electric motor, or other powertrain **102**. The dual pinion and ring architecture allows for a high overall ratio between input and output while avoiding extremely high speeds of the involved gears.

(13) Referring to FIG. 2, a gearset **120** of the gear system **106** is illustrated in schematic form. The gearset **120** receives an input **122**, such as power from the prime mover **104**. The gearset **120** delivers an output **124** such as to the axle **114**. It will be appreciated that between the prime mover **104** and the gearset **120** and/or between the gearset **120** and the axle **114** a number of elements such as links, clutches, brakes, additional gears, and other torque transfer elements may be included to deliver the desired power to the wheels **112** and/or **115** in a desired way.

(14) The gearset **120** may be contained in a housing **130** that may be a type of case, frame, mount, block, or other structure. In the current embodiment, the housing **130** is mounted in the vehicle **100** and provides a fixed ground type structure relative to the gearset **120**. In embodiments, the housing

130 contains a fluid, such as oil for lubricating the gearset **120**. The housing **130** provides a structure on, or within which, to mount the various nonrotating elements of the gearset **120** and to rotatably support the rotating elements.

(15) Referring additionally to FIG. 3, the gearset **120** includes a rotating assembly **128** that, in general, includes a shaft **132**, a pair of counterweights **134** and **136** and a pinion gear unit **138**. The shaft **132** has a bearing section **140**, an eccentric section **142** and a bearing section **144**. The bearing section **140**, the eccentric section **142** and the bearing section **144** may be formed of an integral body or the eccentric section **142** may be separately formed and fixed on a base shaft that includes the bearing sections **140** and **144**. For example, the bearing sections **140** and **144** comprise a base shaft section centered on an axis **170** or may be part of a base shaft that extends through the eccentric section **142** on the axis **170**. The bearing sections **140** and **144** are constrained in bearing assemblies **146** and **148**, respectively. The bearing assemblies **146** and **148** constrain relative motion of the shaft **132** to only the desired rotating motion and to minimize friction during the rotation. The bearing assembly **146** is mounted in an opening **150** of a structural element **152** that may be a part of the housing **130** or that may be fixed to the housing **130**. The bearing assembly **148** is indirectly mounted on the housing **130** through a bearing assembly **126**. As a result, the shaft **132** is rotatable in the housing **130** as driven by the input **122**.

(16) The shaft **132** rotates about the axis **170**. Referring additionally to FIG. 4, the eccentric section **142** has a diameter **160** that is larger than the diameter **162** of the bearing section **140** and that is larger than the diameter **164** of the bearing section **144**. The eccentric section **142** is a cylindrical section fixed to, or formed with, the bearing sections **140** and **144**. As shown in FIG. 2, the bearing sections **140** and **144** have a common center **172** that coincides with the axis **170**. The eccentric section **142** has a center **174** that is offset by an offset distance **176** from the center **172** (and the axis **170**). As a result of the offset distance **176**, the eccentric section **142** has a high side **178** and a low side **180**. Accordingly, when the shaft **132** rotates, the high side **178** travels around the axis **170** providing a circular path for the center **174** around the axis **170**. This also provides a circular path for the eccentric section **142** and anything disposed on it, such as the pinion gear unit **138**. Because the eccentric section **142** has its center positioned off-center relative to the axis **170** of the shaft **132**, as the shaft **132** rotates the center **174** of the eccentric section **142** follows the circular path around the axis **170** as it rotates.

(17) The counterweights **134** and **136** extend from the shaft **132** on the same side of the shaft **132** as the low side **180**. Because of the offset nature of the eccentric section **142**, the weight of the high side **178** as compared to the lower weight of the low side **180**, imparts an imbalance. That imbalance is negated by the counterweights **134** and **136**. In other words, with the counterweights **134** and **136**, the rotating assembly **128** of the shaft **132** and the counterweights **134** and **136** is balanced for rotation. The weight of the pinion gear unit **138** is also considered in the balancing. In embodiments, the eccentric section **142** may have openings to reduce its weight. The counterweights **134** and **136** each have a T-shape with a head **154** and **156**, respectively, the shapes, weights and sizes of which may be tuned for balancing the eccentric section **142**.

(18) Referring additionally to FIG. 5, the pinion gear unit **138** is shown in isolation and in FIGS. 3 and 4 it can be seen that the pinion gear unit **138** is a part of the rotating assembly **128**. The pinion gear unit **138** includes two pinion gears **186** and **188** with sets of gear teeth including teeth **187** and **189** on their respective outside circumferences. In this case, the pinion gears **186** and **188** are part of a stepped pinion type pinion gear unit **138**. The pinion gear **186** is smaller than the pinion gear **188** and the two are formed together as one unit and mounted on the shaft **132** so that they rotate and orbit together. In other embodiments, the pinion gear **186** may be formed as a separate piece from the pinion gear **188** and the two may be mounted on the shaft **132** to rotate and orbit together. A bearing assembly **182**, which may be a needle bearing or another type, is disposed between the eccentric section **142** and the pinion gear unit **138**. In embodiments, two of the bearing assembly **182** may be used. The bearing assembly **182** allows the pinion gear unit **138** to rotate on the

eccentric section **142**.

(19) As shown in FIGS. **2** and **3**, a ring gear **190** is included in the gearset **120**. The ring gear **190** is an internal ring gear with teeth **192** on the inside circumference **194** of the ring-shaped or annular structure. The ring gear **190** and specifically its teeth **192** are designed to mesh with the pinion gear **188**. The ring gear **190** is mounted in the housing **130**. As such, the ring gear **190** is fixed to not rotate relative to the housing **130** and may be referred to as a grounded ring gear.

(20) Another ring gear **210** is included in the gearset **120**. The ring gear **210** is an internal ring gear with teeth **212** on the inside circumference **214** of the ring-shaped or annular structure. The ring gear **210** and specifically its teeth **212** are designed to mesh with the pinion gear **186**. The ring gear **210** has keyways **216** around its outer circumference **218** that engage with keys **220** of a rotatable member referred to as an output cover **230**, or as a cover **230**. The cover **230** is a shell-like member with a shaft **232** attached to a cup **234** that has a flange **236** that extends axially. The flange **236** extends around the radially outside of, and provides a type of cover for, the counterweight **136**. The flange **236** includes the keys **220** within which the ring gear **210** is contained with its keyways **216** engaging with the keys **220** providing a splined type connection. As such, the ring gear **210** and the cover **230** are fixed to each other to rotate together. A pair of snap rings **240** and **242** engage in grooves in the cover **230** to hold the ring gear **210** in position when assembled.

(21) When the shaft **132** is driven to rotate as an input shaft, the pinion gears **186** and **188** orbit around the axis **170** as driven by the eccentric section **142**. As the shaft **132** rotates, the pinion gear **188** travels around the ring gear **190** in its orbit. Specifically, the pinion gear **188** travels around the inside circumference **194** of the ring gear **190**, which causes the pinion gears **186** and **188** to spin on the eccentric section **142**. The pinion gear **186** spins at the same rotational speed as the pinion gear **188**. The pinion gear **186** engages with the ring gear **210**. As a result, the pinion gear **186** causes the ring gear **210** to rotate. In-turn, the rotation of the ring gear **210** imparts rotation to the cover **230** and the shaft **232**, which operates as an output shaft.

(22) The connection between the ring gear **210** and the cover **230** is through the splined connection of the keyways **216** and the keys **220**. Axial loads on the ring gear **210** are transferred to the cover **230**. The axial loads are then transferred to the housing **130** through the bearing assembly **126**, which may be a ball bearing assembly designed to incur axial loading or an axial roller bearing may be employed. Radial loads on the shaft **132** are transferred to the housing **130** through the bearing assembly **148**, the cover **230** and the bearing assembly **126** and also through the bearing assembly **146**.

(23) The gearset **120** may be employed in applications where a high ratio and small size are desirable. The pinion gear unit **138** may be similar to a type of planetary pinion gear that orbits within the ring gears **190** and **210** as a result of the eccentric section **142** of the shaft **132**. The eccentric section **142** may act as a type of planetary carrier for the gear system **106**. High overall ratios may be achieved in a radially compact space. For example, ratios up to and even surpassing 20:1 may be provided. A feature is that higher ratios may be achieved with smaller radial sizing of the gearset **120**. Another feature is that rotational speeds of the pinion gears are maintained at acceptable low levels. Also, since there are only two gear meshes in this gear system **106** there is lower power loss as compared to systems with more meshes. In addition, none of the gears in the gearset **120** require clocking for manufacturing and assembly.

(24) Accordingly, a gear system is provided with only two external-internal gear meshes in the architecture. The two external-internal gear pairs may both be described as planetary-type systems. However, these planetary-type systems only have one gear unit that may be considered similar to a planet gear (external gear) and no sun gear. Two external pinion gears are carried on one shaft and form a stepped pinion type gear. The input shaft has an eccentric section, similar to a cam shaft, that acts as a type of planetary carrier. The rotating assembly containing the two external gears is assembled on the eccentric section of the input shaft with needle bearings between the two parts. One internal ring gear meshes with a relatively larger pinion and is grounded, and another internal

ring gear meshes with a relatively smaller pinion to provide the output of the system. The output ring gear is connected to a cover that includes an output shaft. Depending on the power transfer strategy, this cover may either be connected to a driveline differential or directly splined to an axle half shaft. Counterweights balance the eccentric motion of the system. To assist with assembly, one counterweight may be forged with the input shaft while the other counterweight may be designed to be assembled on the input shaft using press-fit and/or a keyway to ensure correct alignment.

(25) While at least one exemplary embodiment has been presented in the foregoing detailed description, it should be appreciated that a vast number of variations exist. It should also be appreciated that the exemplary embodiment or exemplary embodiments are only examples, and are not intended to limit the scope, applicability, or configuration of the disclosure in any way. Rather, the foregoing detailed description will provide those skilled in the art with a convenient road map for implementing the exemplary embodiment or exemplary embodiments. It should be understood that various changes can be made in the function and arrangement of elements without departing from the scope of the disclosure as set forth in the appended claims and the legal equivalents thereof.

Claims

1. A gear system comprising: a housing; a shaft extending in the housing, the shaft including an eccentric section, wherein the shaft is rotatable about an axis and wherein the eccentric section has a center that circles around the axis when the shaft rotates; a pinion gear unit mounted on the eccentric section, the pinion gear unit having a first set of gear teeth and a second set of gear teeth; a first ring gear engaging with the first set of gear teeth, the first ring gear fixed to the housing; a second ring gear engaging with the second set of gear teeth; a counterweight extending from the shaft to balance the eccentric section, wherein the counterweight has a head with a shape that is T-shaped and that is disposed axially in-line with the first ring gear with a radially outer portion of the head being axially adjacent the first ring gear, wherein the shape is configured to be tuned for balancing the eccentric section; and a rotatable member fixed to the second ring gear, wherein the gear system is configured so that rotation of the shaft at a first speed effects rotation of the rotatable member at a second speed through the shaft, the pinion gear unit, the first ring gear and the second ring gear.
2. The gear system of claim 1, wherein the T-shaped head has a pair of arms that taper.
3. The gear system of claim 1, wherein the rotatable member has a hollow shaft section extending axially along, around and adjacent the shaft, and comprising a first bearing assembly disposed between and contacting both the hollow shaft section of the rotatable member and the housing, and a second bearing assembly disposed between and contacting both the shaft and the hollow shaft section of the rotatable member, wherein axial loads on the second ring gear are transferred to the housing through the second bearing assembly, the rotatable member and the first bearing assembly.
4. The gear system of claim 1, wherein the eccentric section is configured with openings to reduce weight, wherein the openings extend completely through the eccentric section in a direction parallel to the axis.
5. The gear system of claim 1, comprising a needle bearing assembly disposed between the eccentric section and the pinion gear unit so that the pinion gear unit is rotatable on the eccentric section, wherein the needle bearing assembly contacts the eccentric section and the pinion gear unit.
6. The gear system of claim 1, wherein the eccentric section is configured with openings to reduce weight, wherein the openings extend completely through the eccentric section in a direction parallel to the axis, wherein the eccentric section has a high side and a low side, wherein the openings include a first opening that is centered on the high side and that is larger than others of the openings.

7. The gear system of claim 1, wherein the shaft includes a pair of bearing sections that are rotatably supported on the housing, wherein the bearing sections, the first ring gear and the second ring gear are disposed about the axis coaxially and wherein the eccentric section is disposed offset from the axis.

8. The gear system of claim 1, wherein: the housing is mounted in a vehicle; the rotatable member is coupled with a wheel of the vehicle through an axle of the vehicle, the shaft is coupled with a powertrain of the vehicle, and the powertrain drives the wheel through, in order, the shaft, the pinion gear unit, the second ring gear, the rotatable member, and the axle.

9. The gear system of claim 1, comprising no more than two meshing gear pairs, wherein a first meshing gear pair includes the pinion gear unit and the first ring gear and a second meshing gear pair includes the pinion gear unit and the second ring gear.

10. The gear system of claim 1, wherein the pinion gear unit comprises a stepped pinion gear with a first pinion gear and a second pinion gear, wherein the first pinion gear and the second pinion gear are configured to rotate and to orbit about the axis together on the eccentric section.

11. A gear system comprising: a housing; and a gear set in the housing, the gear set including: a shaft extending in the housing, the shaft including a base shaft section and an eccentric section, wherein the shaft is rotatable about an axis, wherein the base shaft section is centered on the axis, and wherein the eccentric section has a center that is offset from the axis, wherein the center circles around the axis when the shaft rotates; a pinion gear unit mounted on the eccentric section, the pinion gear unit having a first pinion gear with a first set of gear teeth and a second pinion gear with a second set of gear teeth; a first ring gear engaging with the first set of gear teeth, the first ring gear fixed to the housing; a second ring gear engaging with the second set of gear teeth; a counterweight extending from the shaft to balance the eccentric section, wherein the counterweight has a head with a shape that is T-shaped and that is disposed axially in-line with the first ring gear with a radially outer portion of the head being axially adjacent the first ring gear, wherein the shape is configured to be tuned for balancing the eccentric section; and a rotatable member fixed to the second ring gear, wherein the gear system is configured so that rotation of the shaft at a first speed effects rotation of the rotatable member at a second speed through the shaft, the pinion gear unit, the first ring gear and the second ring gear.

12. The gear system of claim 11, wherein the T-shaped head has a pair of arms that taper.

13. The gear system of claim 11, wherein the rotatable member has a hollow shaft section extending axially along, around and adjacent the shaft, and comprising a first bearing assembly disposed between the rotatable member and the housing and contacting both the hollow shaft section and the housing, and a second bearing assembly disposed between and contacting both the hollow shaft section of the rotatable member and the shaft, wherein radial loads on the shaft are transferred to the housing through the second bearing assembly, the rotatable member and the first bearing assembly.

14. The gear system of claim 11, wherein the eccentric section is configured with openings to reduce weight, wherein the openings extend completely through the eccentric section in a direction parallel to the axis.

15. The gear system of claim 11, comprising a needle-type bearing assembly disposed between the eccentric section and the pinion gear unit so that the pinion gear unit is rotatable on the eccentric section.

16. The gear system of claim 11, wherein the eccentric section is configured with openings to reduce weight, wherein the openings extend completely through the eccentric section in a direction parallel to the axis, wherein the eccentric section has a high side and a low side, wherein the openings include a first opening that is centered on the high side and that is larger than others of the openings.

17. The gear system of claim 11, wherein the shaft includes a pair of bearing sections as part of the base shaft section so that the shaft is rotatably supported on the housing, wherein the base shaft

section, the first ring gear and the second ring gear are disposed about the axis coaxially and wherein the eccentric section is disposed on a center that is offset from the axis.

18. The gear system of claim 11, wherein: the housing is mounted in a vehicle; the rotatable member is coupled with a wheel of the vehicle through an axle of the vehicle, the shaft is coupled with a powertrain of the vehicle, and the powertrain drives the wheel through, in order, the shaft, the pinion gear unit, the second ring gear, the rotatable member, and the axle.

19. The gear system of claim 11, comprising no more than two meshing gear pairs, wherein a first meshing gear pair includes the first pinion gear and the first ring gear and a second meshing gear pair includes the second pinion gear and the second ring gear.

20. A gear system for a vehicle having a prime mover and an axle, the gear system comprising: a housing fixed to the vehicle; and a gear set in the housing, the gear set including: a shaft coupled with the prime mover and extending in the housing, the shaft including a base shaft section and an eccentric section, wherein the shaft is rotatable about an axis, wherein the base shaft section is centered on the axis, and wherein the eccentric section has a center that is offset from the axis, wherein the center circles around the axis when the shaft rotates; a pinion gear unit mounted on the eccentric section, the pinion gear unit having a first pinion gear with a first set of gear teeth and a second pinion gear with a second set of gear teeth; a first ring gear engaging with the first set of gear teeth, the first ring gear fixed to the housing; a second ring gear engaging with the second set of gear teeth; a counterweight extending from the shaft to balance the eccentric section, wherein the counterweight has a head with a shape that is T-shaped and that is disposed axially in-line with the first ring gear with a radially outer portion of the head being axially adjacent the first ring gear, wherein the shape is configured to be tuned for balancing the eccentric section; and a rotatable member coupled with the axle and fixed to the second ring gear, wherein the eccentric section is configured with openings to reduce weight, wherein the openings extend completely through the eccentric section in a direction parallel to the axis, wherein the eccentric section has a high side and a low side, wherein the openings include a first opening that is centered on the high side and that is larger than others of the openings, wherein the gear system is configured so that rotation of the shaft by the prime mover at a first speed effects rotation of the rotatable member and the axle at a second speed through the shaft, the pinion gear unit, the first ring gear and the second ring gear, wherein the first pinion gear and the second pinion gear are configured to rotate and to orbit about the axis together on the eccentric section.
